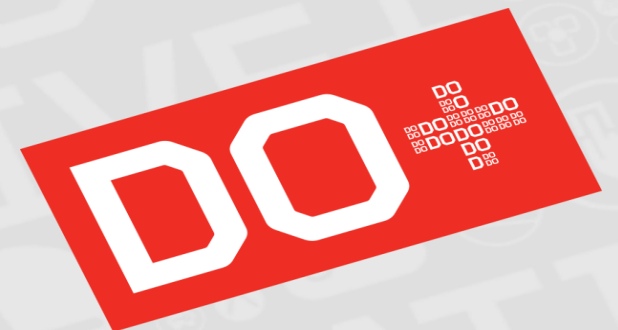


lenovo® FOR
THOSE WHO DO.™

EBG Server Education - xClarity FW Updates & Self-Maintenance

Peter Donovan | 4/15/15



■ Purpose

1. Update firmware on supported hardware

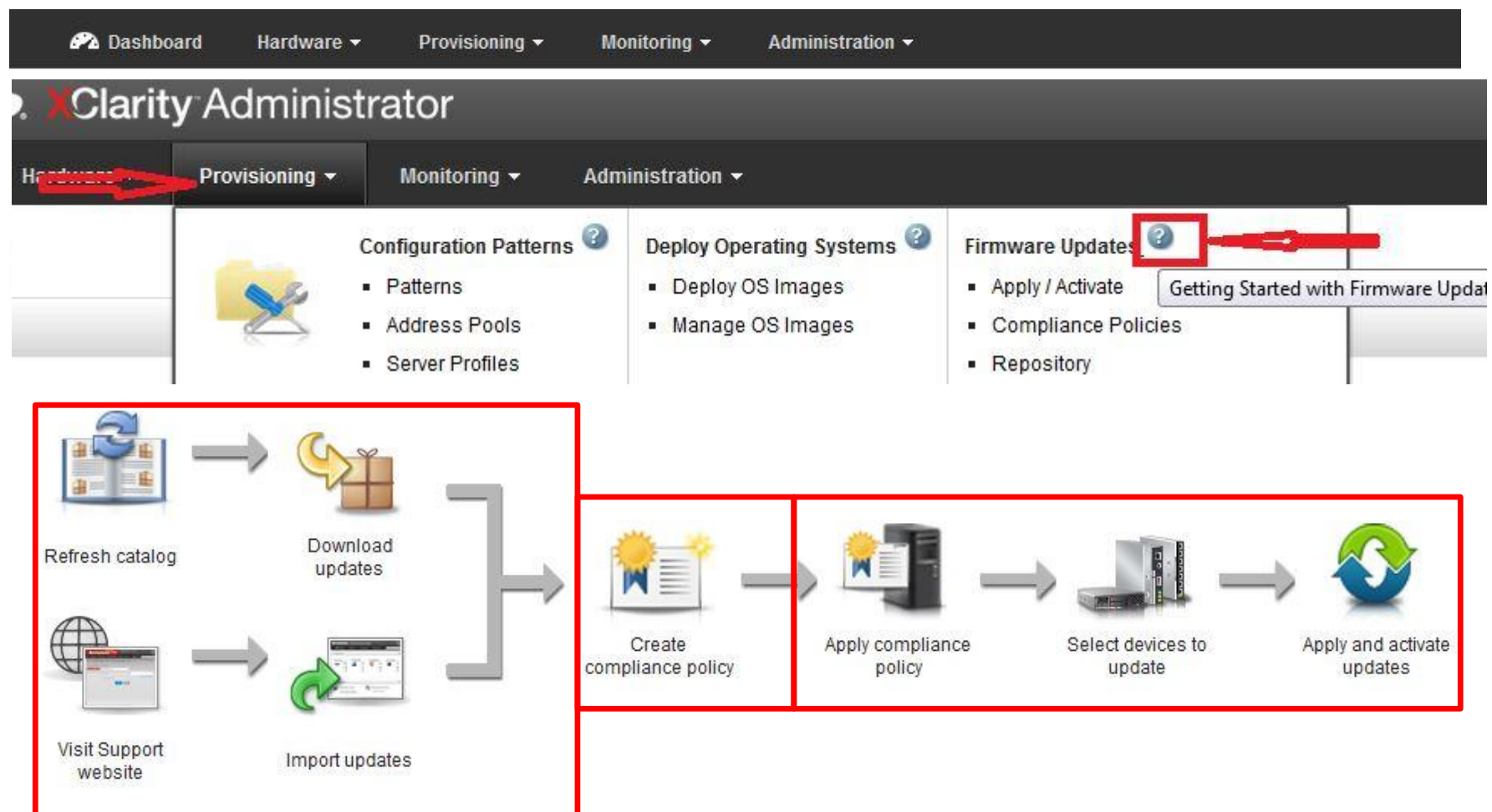
- CMM
- IOM
- System nodes/Rack Systems

2. Update Management Server (Self-Maintenance)

- Two types of Management Server Update (**Patch**, **Supplemental Pack**). Regardless of update type, the same process is used to update Management Server
- **Patch** : Code change to the appliance. Changes minor version of Management Server
- **Supplemental Pack** - Data content changes, such as new Update Repository/Policy

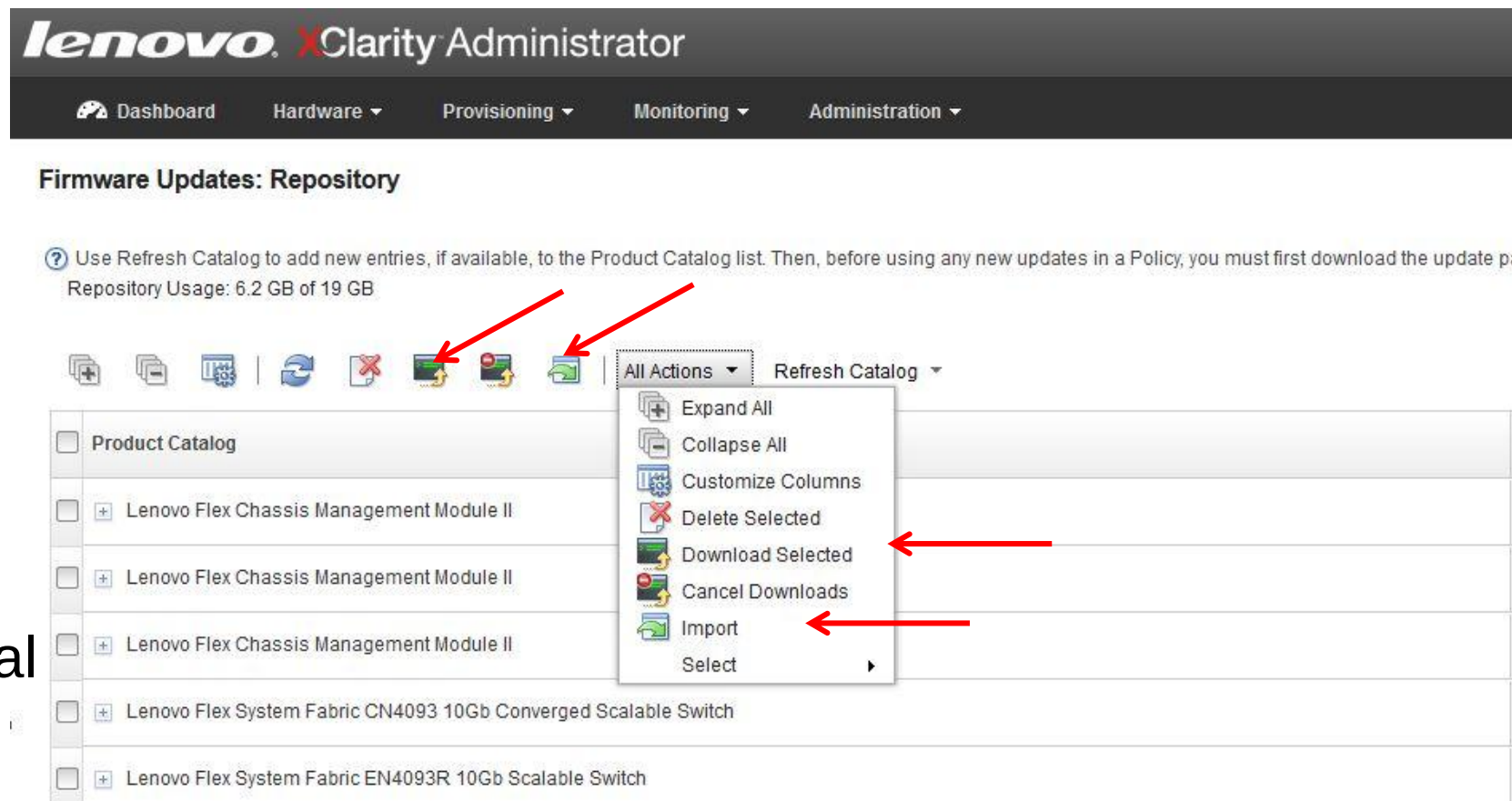
Compliance Policy Based

- Repository
- Policy
- Apply/Activate



Repository

- Preloaded
- Download updates
 - Electronic downloads
 - Import after downloading from IBM FixCentral
- Refresh Catalog
- Download Selected
- Or Import from FixCentral (linux updates)



Repository View

- Updates grouped by MT or IOM type
- System updates further grouped by hardware component

Firmware Updates: Repository

Use Refresh Catalog to add new entries, if available
Repository Usage: 6.2 GB of 19 GB

Product Catalog

- Lenovo Flex System x240 Compute Node
 - ServerRAID M5115 SAS/SATA Controller
 - ServerRAID M5115 SAS/SATA Controller Firmware Update for Linux
Invgy_fw_sraidmr_5115-23.33.0-0018_linux_32-64.txt
- Emulex HBA (LPe1205/LPe1200) Firmware Update for Linux
 - Emulex HBA (LPe1205/LPe1200) Firmware Update for Linux
elx-invgy_fw_fc_15a-2.02x11-16
- IMM2
 - Integrated Management Module
Invgy_fw_imm2_tcoo08r-1.61_a
 - Integrated Management Module
Invgy_fw_imm2_tcoo07r-1.61_a
- Online Broadcom NetXtreme II Linux
- Online Broadcom NetXtreme II

Release Notes

File Name: Invgy_fw_sraidmr_5115-23.33.0-0018_linux_32-64.txt

ReadMe Change Log

ServerRAID m5115 SAS/SATA Controller Firmware Update for Linux

Adapters Supported: ServerRAID M5115 SAS/SATA Controller (90Y4390)

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	Release Notes	Policy Usage	Severity
MB		 In Use	 Suggested
2 MB		 In Use	 Suggested
		 In Use	 Initial Release
MB		 In Use	 Initial Release

Lenovo XClarity Update Compliance Policy

Policy

- Preloaded
- Create
- Copy
- Edit
- Delete

Firmware Updates: Compliance Policies

? Compliance Policy allows you to create or modify a policy based on the acquired updates in the Firmware Repository.



☐ Compliance Policy Name	Usage Status	Compliance Policy Origin
☐ BLUE-FLEX-DEV-2015-04-12	Assigned	Predefined
☐ RED-FLEX-DEV-2015-04-12	Assigned	Predefined
☒ DEFAULT-2015-04-07	Assigned	Predefined
☐ RED-RACK-DEV-2015-04-12	Not Assigned	Predefined
☐ Copy-DEFAULT-2015-04-07	Not Assigned	User Defined
☐ RED-RACK-DEV-2015-04-12	Not Assigned	Predefined

Edit Compliance Policy

- Organized by MT
- Compliance Rule
- Updates by Component

Edit Policy

Name: Copy-DEFAULT-2015-04-07

Description: User created Policy by copying from existing Policy

System Type	Compliance Target	Compliance Rule	Delete
Lenovo Flex System Fabric CR4093 10Gb Converged Scalable Switch - 0x0130-0x0027	8.2.1.0 / 0802(2015-03-23)	Flag if Downlevel	
Lenovo Flex System Fabric EN4093R 10Gb Scalable Switch - 0x0130-0x0028	8.2.1.0 / 0802(2015-03-23)	Flag if Downlevel	
Lenovo Flex System Fabric S14093 System Interconnect Module - 0x0130-0x0029	8.2.1.0 / 0802(2015-03-23)	Flag if Downlevel	
Lenovo Flex System S14091 10Gb System Interconnect Module - 0x0130-0x0026	8.2.1.0 / 0802(2015-03-23)	No Flag	
Lenovo Flex System x240 Compute Node - 2588	Custom	Flag if Not Exact Match	
Integrated Management Module 2 (IMM2) Update	1.61 / TCO08R(2015-04-04)(Default)		
Integrated Management Module 2 (IMM2) Update (Standby)	Do Not Update		
Flex System x240 UEFI Flash Update	1.10 / A3E104I(2015-03-30)(Default)		
Flex System x240 UEFI Flash Update (Standby)	1.10 / A3E103I(2015-03-29)		
Lenovo Dynamic System Analysis (DSA) - Preboot Embedded	10.1 / DSALA60(2015-04-02)(Default)		
Emulex HBA (LPe1205/LPe1200x) Firmware Update for Linux	2.02x11-16 / (2015-03-27)(Default)		

+ Add new system

Compliance Policy after editing

Firmware Updates: Compliance Policies

? Compliance Policy allows you to create or modify a policy based on the acquired updates in the Firmware Repository.



☐	Compliance Policy Name	Usage Status	Compliance Policy Ori... ▲	Last Modified	Description	
☐	BLUE-FLEX-DEV-2015-04-12	Assigned	Predefined	2015-04-12 17:30:06		
☐	RED-FLEX-DEV-2015-04-12	Assigned	Predefined	2015-04-12 17:43:40		
☐	DEFAULT-2015-04-07	Assigned	Predefined	2015-04-07 10:44:00		
☐	RED-RACK-DEV-2015-04-12	Not Assigned	Predefined	2015-04-12 17:09:11		
☐	custom-2015-04-07	Not Assigned	User Defined	This policy was edited by USERI...	User created Policy by copying from existing Policy	

Lenovo XClarity Update Apply/Activate

Apply Update

Update Summary
Select your Update Rule and review your updates. Then click Perform Update.

Note: The update job will run in the background and might take several minutes to complete. Updates are performed as a job. You can go to the Jobs page to view the status of the job as it progresses.

* Update Rule: ?

* Activation Rule: ?

☐ Attempt to update components already in compliance

Filter

System	Chassis / Bay	Installed Version	Compliance Target
☐ x240 Blue 10 10.243.13.197	BEEP, Bay 10		
IMM2 (Primary)	BEEP, Bay 10	4.90 / 1A0065E	1A0065I-4.90 / 1A0065I ibm_fw_imm2_1a0065i-4.9...
UEFI (Primary)	BEEP, Bay 10	1.60 / B2E145GUS	B2E145H-1.60 / B2E145H ibm_fw_uefi_b2e145h-1.60...

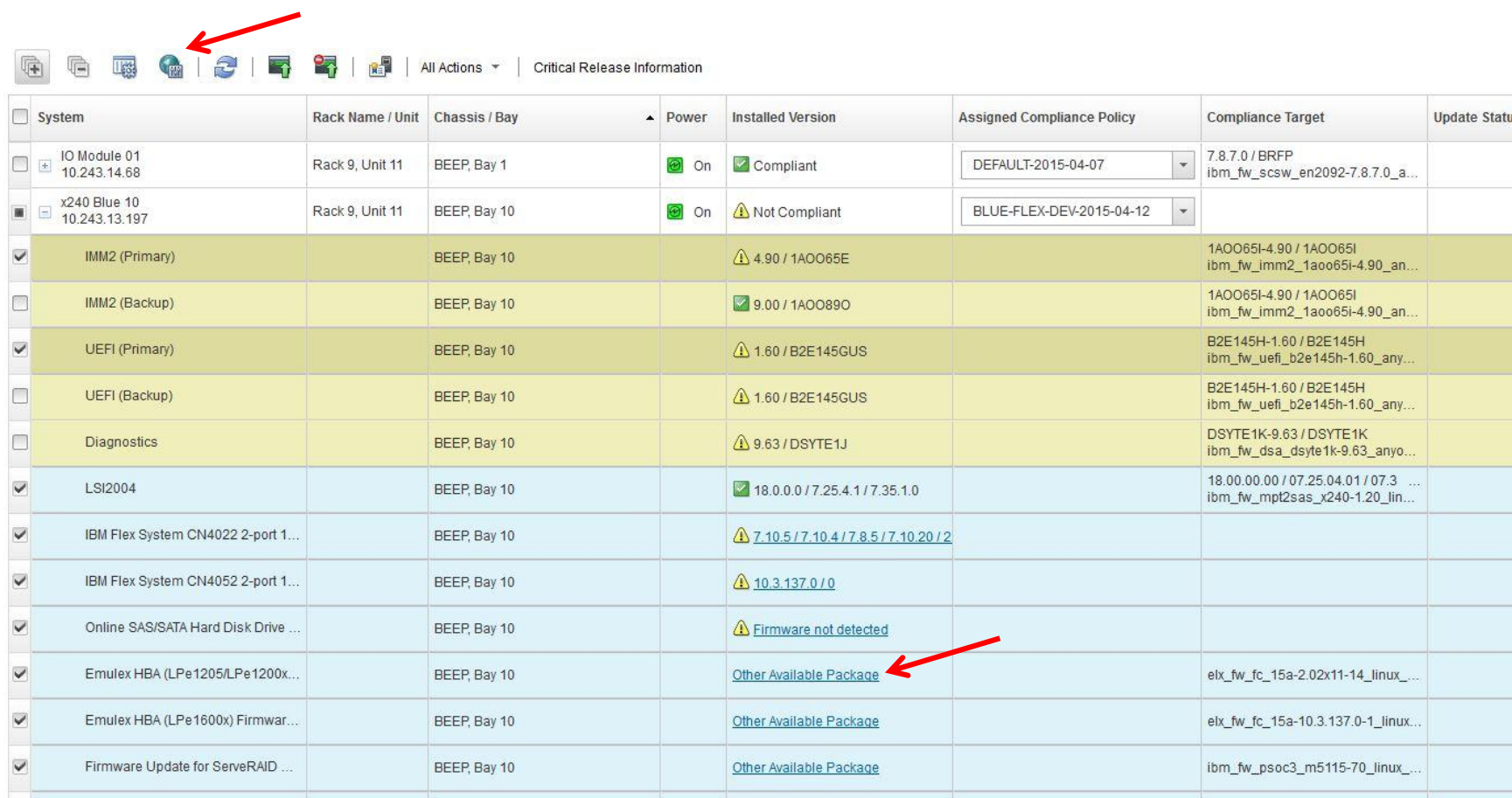
Apply Summary

- Multiple end points can be selected
 - Update are automatically ordered and queued as appropriate
 - Priority: 1. CMM, 2. IOM (only one active on each chassis), 3 System
 - Queueing: Throttle limit is automatically set to 20 active end points
- Update Rule
 - Stop all updates on error
 - Continue on error
 - Continue to next system on error
- Activation Rule
 - Immediate activation : Reboot System to
 - Delayed activation : Only valid for System updates. Out of band updates applied, on next reboot take effect; **Reboot of system** applies Option and Adapter. Limit of 20 end point at one time.

Lenovo XClarity Update Apply/Activate

Apply/Activate Compliance Rule

- Compliance rule
 - Flag if Downlevel
 - Flag if Not Exact Match
 - No Flag
 - Set by Compliance Policy
- Installed version
- Target Compliance
- Enhance Support for Down-level Devices
 - Determines if “Other Available Packages” are applied



System	Rack Name / Unit	Chassis / Bay	Power	Installed Version	Assigned Compliance Policy	Compliance Target	Update Status
IO Module 01 10.243.14.68	Rack 9, Unit 11	BEEP, Bay 1	On	Compliant	DEFAULT-2015-04-07	7.8.7.0 / BRFP ibm_fw_scsww_en2092-7.8.7.0_a...	
x240 Blue 10 10.243.13.197	Rack 9, Unit 11	BEEP, Bay 10	On	Not Compliant	BLUE-FLEX-DEV-2015-04-12		
IMM2 (Primary)		BEEP, Bay 10		4.90 / 1A0065E		1A0065I-4.90 / 1A0065I ibm_fw_imm2_1a0065i-4.90_an...	
IMM2 (Backup)		BEEP, Bay 10		9.00 / 1A00890		1A0065I-4.90 / 1A0065I ibm_fw_imm2_1a0065i-4.90_an...	
UEFI (Primary)		BEEP, Bay 10		1.60 / B2E145GUS		B2E145H-1.60 / B2E145H ibm_fw_uefi_b2e145h-1.60_any...	
UEFI (Backup)		BEEP, Bay 10		1.60 / B2E145GUS		B2E145H-1.60 / B2E145H ibm_fw_uefi_b2e145h-1.60_any...	
Diagnostics		BEEP, Bay 10		9.63 / DSYTE1J		DSYTE1K-9.63 / DSYTE1K ibm_fw_dsa_dsyte1k-9.63_ano...	
LSI2004		BEEP, Bay 10		18.0.0.0 / 7.25.4.1 / 7.35.1.0		18.00.00.00 / 07.25.04.01 / 07.3 ... ibm_fw_mpt2sas_x240-1.20_lin...	
IBM Flex System CN4022 2-port 1...		BEEP, Bay 10		7.10.5 / 7.10.4 / 7.8.5 / 7.10.20 / 2			
IBM Flex System CN4052 2-port 1...		BEEP, Bay 10		10.3.137.0 / 0			
Online SAS/SATA Hard Disk Drive ...		BEEP, Bay 10		Firmware not detected			
Emulex HBA (LPe1205/LPe1200x...		BEEP, Bay 10		Other Available Package		elx_fw_fc_15a-2.02x11-14_linux_...	
Emulex HBA (LPe1600x) Firmwar...		BEEP, Bay 10		Other Available Package		elx_fw_fc_15a-10.3.137.0-1_linux...	
Firmware Update for ServeRAID ...		BEEP, Bay 10		Other Available Package		ibm_fw_psoc3_m5115-70_linux_...	

Apply Update: Running Update View

--	--	--	--	--	--	--	--	--	--	--	--	--	--

Apply Update: Running Update View (Maintenance Mode)

Critical Release Information							
All Actions							
☐	System	Rack Name / Unit	Chassis / Bay ▲	Power	Installed Version	Assigned Co...	Update Status
☐	x240 Blue 10 10.243.13.197	Rack 9, Unit 11	BEEP, Bay 10	On	Not Compliant	BLUE-FLEX-D	Failed
☐	IMM2 (Primary)		BEEP, Bay 10		4.90 / 1A0O65I	1A0O65I-4.90 / 1A0O65I ibm_fw_imm2_1aoo65i-4.90_an...	Complete
☐	IMM2 (Backup)		BEEP, Bay 10		9.00 / 1A0O89O	1A0O65I-4.90 / 1A0O65I ibm_fw_imm2_1aoo65i-4.90_an...	
☐	UEFI (Primary)		BEEP, Bay 10		1.60 / B2E145HUS	B2E145H-1.60 / B2E145H ibm_fw_uefi_b2e145h-1.60_any...	Complete
☐	UEFI (Backup)		BEEP, Bay 10		1.60 / B2E145GUS	B2E145H-1.60 / B2E145H ibm_fw_uefi_b2e145h-1.60_any...	
☐	Diagnostics		BEEP, Bay 10		9.63 / DSYTE1J	DSYTE1K-9.63 / DSYTE1K ibm_fw_dsa_dsyte1k-9.63_ano...	
☐	LSI2004		BEEP, Bay 10		18.0.0.0 / 7.25.4.1 / 7.3	18.00.00.00 / 07.25.04.01 / 07.3 ibm_fw_mpt2sas_x240-1.20_lin...	Skipped (already compliant)
☐	IBM Flex System CN...		BEEP, Bay 10		7.10.5 / 7.10.4 / 7.8.5 /		
☐	IBM Flex System CN...		BEEP, Bay 10		10.3.137.0 / 0		
☐	Online SAS/SATA Ha...		BEEP, Bay 10		Firmware not detected		
☐	Emulex HBA (LPe12...		BEEP, Bay 10		Other Available Package	elx_fw_fc_15a-2.02x11-14_linux_...	Failed
☐	Emulex HBA (LPe16...		BEEP, Bay 10		Other Available Package	elx_fw_fc_15a-10.3.137.0-1_linux...	Failed

Apply Update Result

	System	Rack Name / Unit	Chassis / Bay ▲	Power	Installed Version
☐	x240 Red 01 10.243.12.96	Rack 9, Unit 11	BEEP, Bay 1	Off	No Compliance P
☐	BEEP-BEEP 10.243.1.139	Rack 9, Unit 11	BEEP, Bay 1	On	1.0.0 / 1AON010
☐	IO Module 01 10.243.14.68	Rack 9, Unit 11	BEEP, Bay 1	On	Compliant
☐	x240 Blue 10 10.243.13.197	Rack 9, Unit 11	BEEP, Bay 10	On	Not Compliant
☐	IMM2 (Primary)		BEEP, Bay 10		4.90 / 1AOO65I
☐	IMM2 (Backup)		BEEP, Bay 10		9.00 / 1AOO89O
☐	UEFI (Primary)		BEEP, Bay 10		1.60 / B2E145HUS
☐	UEFI (Backup)		BEEP, Bay 10		1.60 / B2E145GUS
☐	Diagnostics		BEEP, Bay 10		9.63 / DSYTE1J
☐	LSI2004		BEEP, Bay 10		18.0.0.0 / 7.25.4.1

All Actions ▼ | Critical Release Information

Update Status Details

3 Messages [Hide Details](#)

An error occurred while performing firmware update to the Emulex HBA (LPe1205/LPe1200x) Firmware Update for Linux inband component. FQXHMUP4709L [Hide Details](#) ×

The ToolsCenter update utility indicated a generic failure.

Retry the operation. If the problem persists, please contact Customer Support.

An error occurred while performing firmware update to the E... FQXHMUP4709L [Show Details](#) ×

An error occurred while performing firmware update to the Fir... FQXHMUP4709L [Show Details](#) ×

Close

Jobs

?

Jobs are longer running tasks performed against one or more target systems. After selecting a job, you can choose to cancel it, delete it, or obtain details

Calendar

Gears

Stop

Cancel

Refresh

All Actions

Job

Firmware Updates

Success

Success

Success

Success

Success

Skipped (already compliant)

Failed

Failed

Failed

Hardware not present

Hardware not present

Success

Success

Success

▼ Summary:

Name:

Firmware Updates

Time Started:

April 14, 2015 at 06:30:10

Overall Status:

✖

Stopped With Error

Target:

BEEP: x240 Blue 10: Bay 10

Created By:

USERID

▼ Target Results: With Errors: 3 Running: 0 Completed: 11

Target	Message
BEEP: x240 Blue 10: Bay 10: System Prerequisites	Success
BEEP: x240 Blue 10: Bay 10: IMM2 (Primary)	Success
BEEP: x240 Blue 10: Bay 10: UEFI (Primary)	Success

Close

Lenovo XClarity Update Management Server

lenovo. XClarity Administrator

Dashboard Hardware Provisioning Monitoring Administration

Update Management Server

Update the management server software to the latest level.

[Update Management Server: Getting Started](#)

Before updating, make sure that you:

- Back up the management server. [Learn more](#)
- Check the job log to make sure that there are no jobs currently running.

Lenovo® XClarity Administrator

Version:	1.0.0
Last Updated:	04-06-2015-03:18
Repository Usage:	0 KB of 20 GB



All Actions

-  Check for Available Updates
-  Import

Update Name	Version	Release Date
No updates currently available		

Update Management Server

- Two types of Management Server Update (**Patch, Supplemental Pack**). Regardless of update type, the same process is used to update Management Server
- **Patch** : Code change to the appliance. Changes minor version of Management Server
- **Supplemental Pack** - Data content changes, such as new Update Repository/Policy

Troubleshooting

■ Repository

- Electronic Download reports error: **Solve:** Check network and Proxy settings
- Updates not imported: **Solve:** XML file is required, if others are missing then update is partially downloaded. Reimport all update files before applying to Policy

■ Policy

- Policy is organized by MT. For user defined Policies ensure you include MT for the System name you are managing

■ Apply/Activate

- Cannot apply Policy: **Solve:** Ensure policy includes support for CMM, IOM, or System. Internal Lenovo users may have test signed end points, in that case ensure that Policy exists for test signed and production signed firmware.

Troubleshooting

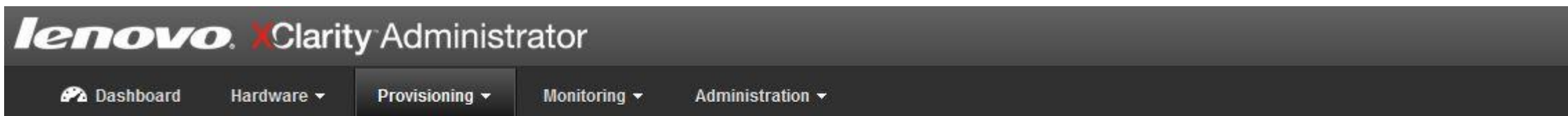
■ Apply/Activate

- System component doesn't have target update (for example a RAID adapter component). **Solve:** Target Compliance check requires Agentless support on adapters, apply all updates to system end point.
- Update fails. **Solve:** For CMM, IOM and core fw (IMM, pDSA, UEFI) after some time retry update. If update still fails reboot or restart end point.
- Adapter/option Update fails. **Solve:** Retry update. If update is not supported on end point, then update failure may be ignored.
- General update failures. **Solve:** View online help for latest information.

Key Technologies

- Bare Metal Update (BMU)
 - Maintenance Mode OS used to apply Adapter and Option Updates
- Agentless
 - Used to gather inventory for Adapter and Option compliance check

Critical Update Information



Firmware Updates: Apply / Activate

? To update firmware for a device, assign a compliance policy and select Perform Updates.

Icons for various actions: Add, Edit, Settings, Refresh, Power On, Power Off, and a dropdown menu labeled 'All Actions'. The 'Critical Release Information' option is highlighted with a red box.

☐	System	Rack Name / Unit	Chassis / Bay	Power	Installed Version	Assigned Compliance Policy	Comp
☐	x240 Red 01 10.243.12.96	Rack 9, Unit 11	BEEP, Bay 1	Off	No Compliance Policy Set	No applicable policies	
☐	BEEP-BEEP 10.243.1.139	Rack 9, Unit 11	BEEP, Bay 1	On	1.0.0 / 1AON010	No applicable policies	
☐	IO Module 01 10.243.14.68	Rack 9, Unit 11	BEEP, Bay 1	On	Compliant	DEFAULT-2015-04-04	7.8.7. ibm_f
☐	x240 Blue 10 10.243.12.107	Rack 9, Unit 11	BEEP, Bay 10	Off	No Compliance Policy Set	No applicable policies	